

Three dental crowns are shown against a dark, textured background. The crown in the center is a full-coverage crown with a metal core and a ceramic veneer. The two crowns on either side are also full-coverage crowns with ceramic veneers. The crowns are arranged in a slightly overlapping manner.

Dental Ceramics II

Dental Medicine Faculty

Biomaterials

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A-Classification: (Craig Textbook 2020)

I-Classification according to application:(Craig Textbook 2020)

Ceramics have two major applications in dentistry:

- (1)ceramics for metal-ceramic crowns and fixed dental prostheses (FDPs),
- (2) all ceramic materials for crowns , inlays, onlays, veneers, and FDPs.

In addition, ceramic orthodontic brackets, dental implant abutments, and ceramic denture teeth (**rare**) are available.



Cutaways of all-ceramic crown (*left*) and porcelain fused to metal crown (*right*). (Courtesy of Dr. Charles Mark Malloy, Portland, Oregon.)

<https://pocketdentistry.com/11-restorative-materials-ceramics/>

II- Classification by Fabrication Method

Different fabrication methods is summarized in following figure.

1- **Metal ceramic restorations**: The most common fabrication technique for metal ceramic restorations is  *sintering*.

2- **All-ceramic restorations** :can also be produced by sintering, **but** they have a wider range of other processing techniques, including slip casting, heat pressing, and CAD/CAM machining.

Some of these techniques, such as machining and heat pressing, can also be combined to produce the final restoration.

B-GENERAL APPLICATIONS OF CERAMICS IN PROSTHETIC DENTISTRY (Craig Textbook 2020)

Metal-Ceramic Crowns and Fixed Dental Prostheses

Ceramic is used as the veneering material in metal-ceramic crowns and FDPs. This development was the result of successfully matching the coefficients of thermal expansion of porcelain with that of metal alloys by  a proper metal-ceramic bond.



All-Ceramic Crowns, Inlays, Onlays, and Veneers

Practical

- Machining and pressing gradually replacing slip casting

A- Ceramic inlays and onlays are replacing indirect composites due to better abrasion resistance. (ADV.)

However, ceramics have more difficult occlusal adjustments and if are not well polished can cause wear of opposing teeth (DISADV.)



B-Ceramic veneers : layer of ceramic bonded to facial prepared surfaces of anterior teeth for cosmetic reasons.

Practical

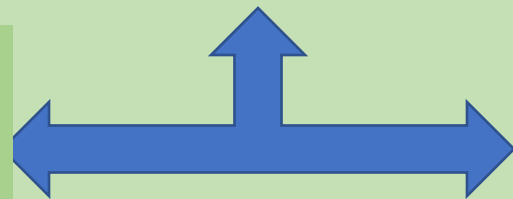
1- MATERIAL:- in the past were made from sintered feldspathic porcelain

- But now are made of leucite reinforced or lithium disilicate ceramics, fabricated by machining or pressing



2- Cementation of ceramic veneers **by special resin composite cements:**

The bonding surface of the ceramic is etched with 5% to 9% hydrofluoric acid gel and treated with a silane coupling agent.



enamel prepared surface is etched by phosphoric acid and bonding agent

Practical

https://www.researchgate.net/publication/51632609_Bonding_All-Ceramic_Restorations_with_Two_Resins_Cement_Techniques_A_Clinical_Report_of_Three-Year_Follow-Up/figures?lo=1



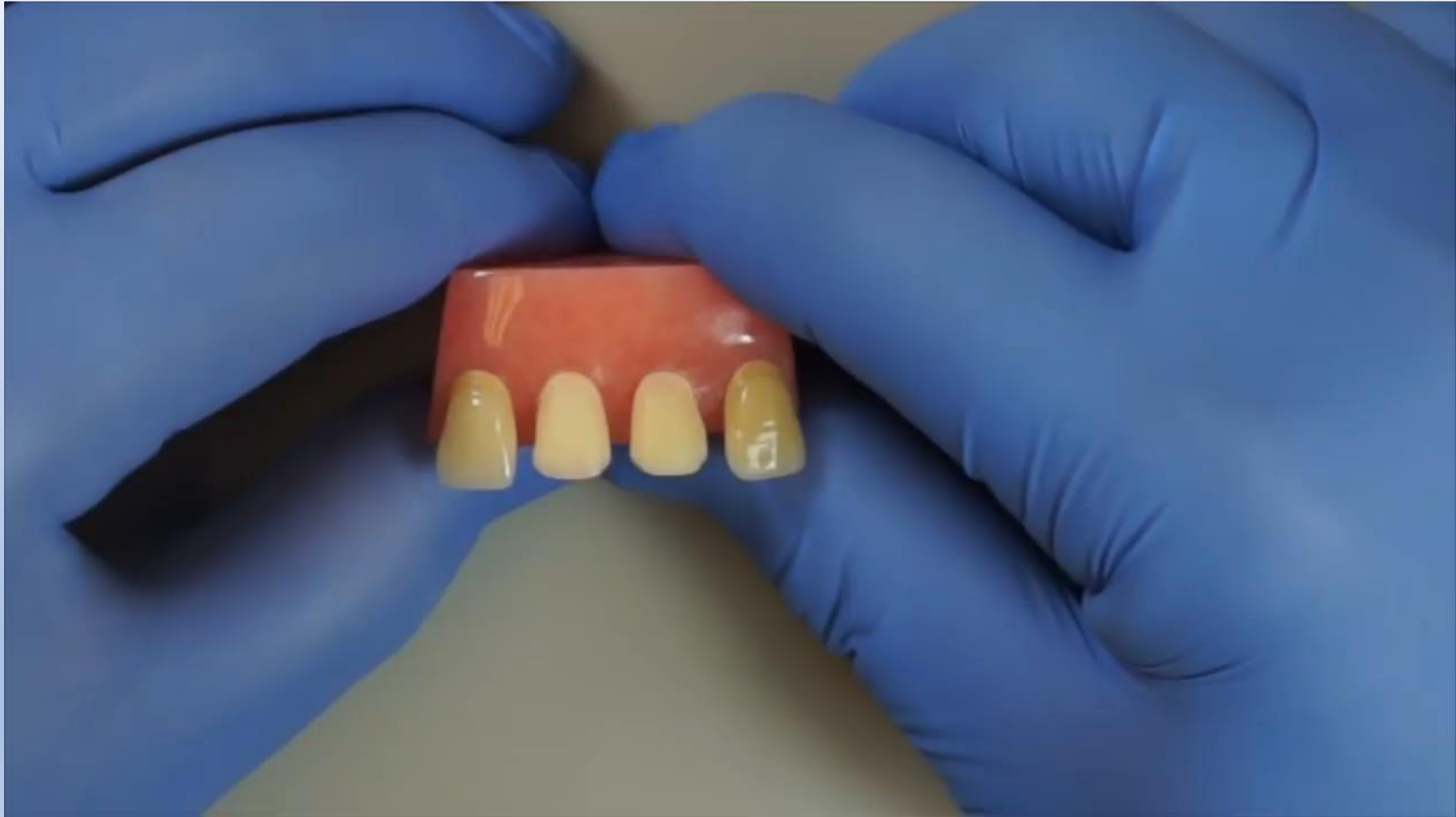
Ceramic crowns and laminate veneers ready for cementation

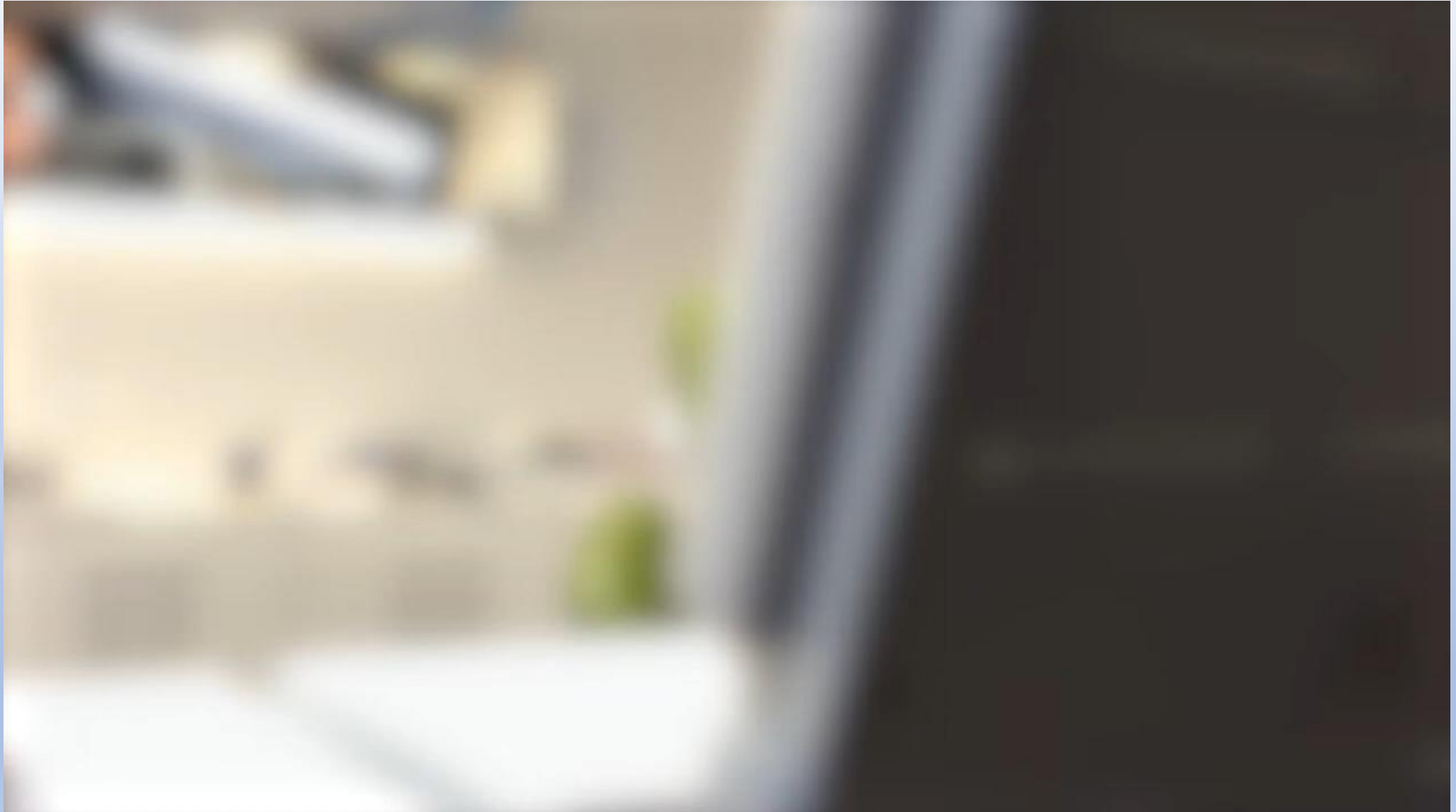


Hydrofluoric acid etching of the laminate.

https://www.researchgate.net/publication/51604787_Esthetic_rehabilitation_of_worn_anterior_teeth_with_thin_porcelain_laminate_veneers/figures?lo=1

Practical





C- Properties (Craig Textbook 2020)

I- Mechanical properties:

A-Toughening Mechanisms

Toughening mechanisms for glasses and ceramics are either:

1- “Built-in” (Intrinsic) :depending on chemical composition or crystalline phase
e.g.: toughening transformation or crystalline reinforcement

or

2- Extrinsic : by external factors as glazing, Tempering or chemical strengthening

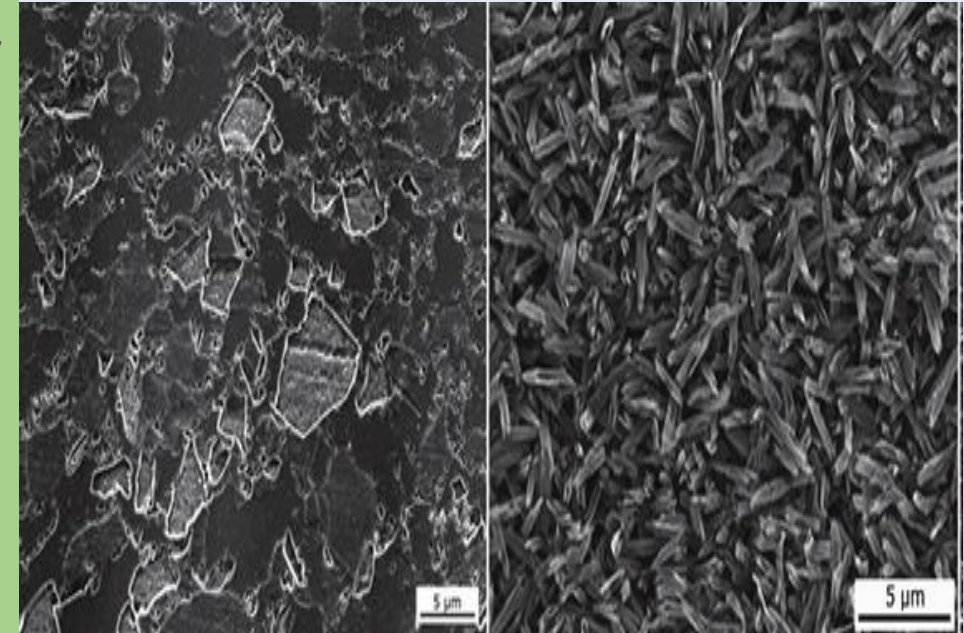
Intrinsic

I-Principle of toughening by crystalline reinforcement : increase resistance to crack propagation by dispersed high toughness crystals

Crystals can act as crack deflectors, when they have $CTE > CTE$ of surrounding glass matrix $\longrightarrow$ placing them under **TANGENTIAL COMPRESSIVE STRESS**, why?



Because crystalline phase shrinks more than surrounding glassy phase matrix on cooling



https://www.researchgate.net/figure/Microstructure-of-the-investigated-materials-a-Leucite-reinforced-glass-ceramic-IEM-b_fig1_221909436

II-Stress-induced transformation toughening

Stress-induced transformation toughening is obtained, for example, in ceramics consisting of *partially stabilized tetragonal zirconia*

Zirconia (ZrO_2) exists under different crystallographic forms.

The monoclinic form is stable at all temperatures below 1170°C .

The tetragonal form is stable between 1170° and up to 2370°C .

The transformation from the tetragonal to the monoclinic form by cooling causes a volume *increase* of the unit cell.



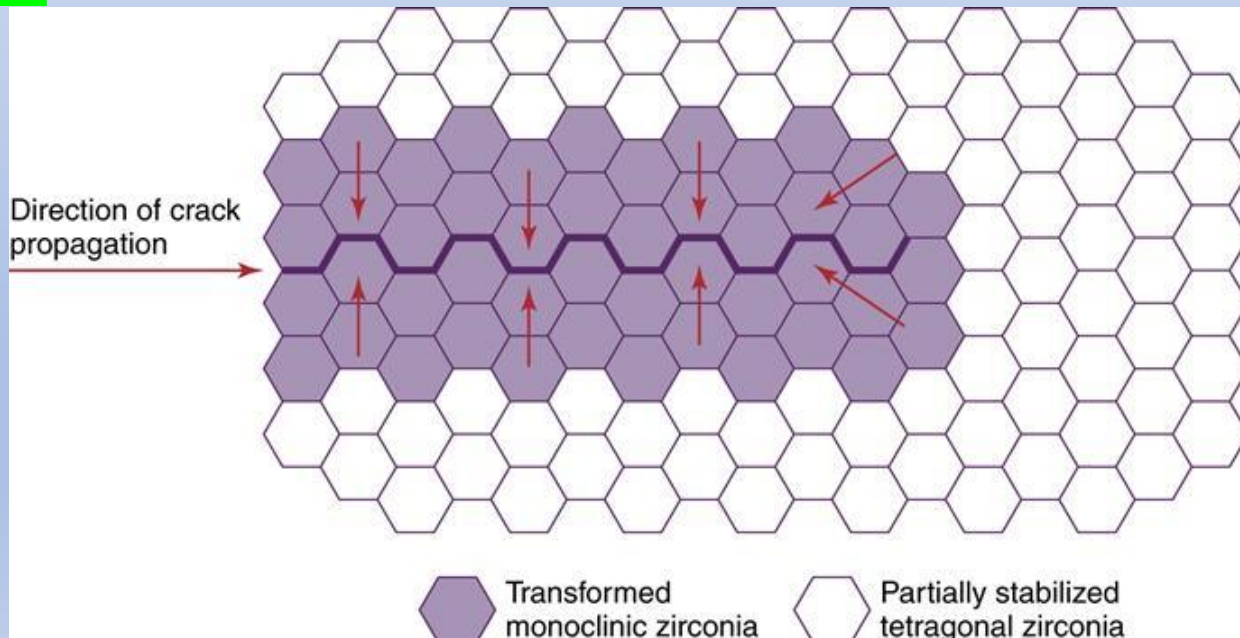
So, pure unalloyed zirconia cannot be obtained at room temperature; as the zirconia would shatter upon cooling due to the transformation and volume increase WHY?

Because Tetragonal changed to higher volume monoclinic form.

How to overcome this problem????

The tetragonal form can *be partially stabilized* to room temperature by doping with different oxides, such as yttrium oxide (Y_2O_3) or cerium oxide (CeO_2).

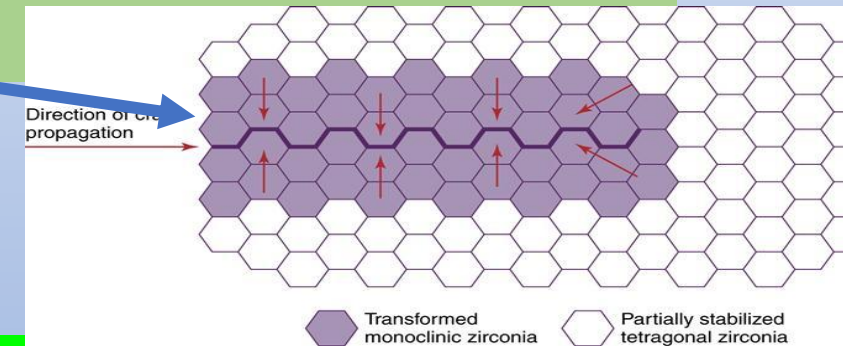
e.g. Zirconia-based dental ceramics produced by machining followed by sintering at high temperatures consist of: tetragonal zirconia polycrystals, partially stabilized by 3 mole percent yttrium (3Y-TZP).



Advantages of metastability of zirconia ??????????????

This partial stabilization or metastability of the tetragonal phase allows the transformation from tetragonal to monoclinic to occur when ?? When external stresses are applied.

This transformation is *stress induced* and is accompanied by a volume increase + compressive stresses at the crack tip leading to closing and arrest of the crack in the transformed zone



Transformation toughening leads to the excellent mechanical properties of 3Y-TZP.

II- Extrinsic

Glazing: is an extrinsic strengthening method and is the final step in the fabrication of metal-ceramic restorations.

- A low-expansion glass called *glaze* can be applied to the surface of the ceramic, then fired to high temperature. Upon cooling, this glaze layer is placed under compression from the greater contraction of the underlying ceramic.

- This layer reduces depth and width of the surface flaws, thereby improving the overall resistance of the ceramic to crack propagation.



An aerial photograph of a long, multi-lane highway bridge spanning a body of green water. The bridge has several lanes in each direction, with white lane markings. Several vehicles, including cars and trucks, are visible traveling across the bridge. The water is a vibrant green color with visible ripples. The text "Thank you" is overlaid on the image in a white, italicized font, underlined.

Thank you